Adaptive rewiring produces sharp finite-size bimodality in a conviction-weighted competitive network

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Abstract

We study a three-species competitive model in which each node carries a continuous three-channel state and interactions favor the locally dominant species. The order parameter ρ_b is the fraction of edges connecting nodes of different dominant species, with low ρ_b corresponding to species-segregated order and high ρ_b to a well-mixed disordered state. We ask how coupling these local dynamics to adaptive rewiring changes the finite-size behavior of the mutation-driven order–disorder transition. At $L = 128$, the fixed toroidal lattice shows a smooth crossover-like loss of order, while the adaptive network shows a sharp jump from $\rho_b \approx 0.17$ to ≈ 0.48 across $\mu \in [0.35, 0.355]$ together with bimodal per-seed distributions. Finite-size Binder analysis disfavors conventional first-order scaling over the tested sizes: the Binder dip weakens from $L = 128$ to $L = 256$, the bimodal window narrows, and the $L = 384$ result is consistent with this trend but limited by equilibration. The data are consistent with a sharp continuous transition over the tested range, although a weakly first-order transition with crossover length beyond $L = 384$ cannot be excluded. Degree-cap controls indicate that degree headroom contributes to the sharpening, but the mechanism remains partly confounded by reinforcement-firing and formation-rate effects. Adaptive topology thus emerges as a strong finite-size sharpening mechanism in this conviction-weighted competitive model. The asymptotic order of the transition remains open.

1. Introduction

Multi-species competition on lattices is a well-studied route to spatial pattern formation. Models of cyclic and non-cyclic competition produce phase transitions between ordered (species-segregated) and disordered (well-mixed) states whose universality depends on lattice geometry, number of species, and the form of the local dynamics [1, 2, 3]. On fixed regular lattices, these transitions are typically continuous.

Real populations, however, do not interact on static topologies. Individuals form and break social, spatial, or ecological links in response to their own state and the states of their neighbors—a coevolutionary feedback between dynamics *on* the network and dynamics *of* the network [4]. A growing body of work shows that this feedback can qualitatively change collective behavior, including the order of phase transitions. Holme and Newman [5] and Vazquez et al. [6] demonstrated that adaptive rewiring converts the continuous fragmentation transition in the voter model to a discontinuous one. Gross et al. [7] found analogous sharpening in adaptive epidemic models, and Demirel et al. [8] studied adaptive networks with cyclic dominance, showing that coevolution of topology and strategy leads to spontaneous network fragmentation.

Here we ask whether topological coevolution changes the order of the order–disorder transition in a conviction-weighted multi-species competition model. The transition we study is between a low-boundary-density, species-segregated phase and a high-boundary-density, well-mixed phase, driven by the mutation probability μ (the boundary density ρ_b is the fraction of edges connecting nodes of different dominant species; defined precisely in Section 2). We further ask what specific feature of the rewiring drives any change in transition character. Our system differs from prior adaptive-network studies in three respects: (i) node states are continuous three-channel vectors with a *dominance margin* (the difference between the largest and second-largest channel) that determines who concedes in pairwise conflicts—a soft, "conviction-weighted" analogue of majority or voter dynamics; whether the continuous margin is essential to the phenomenology, as opposed to a discrete-state simplification, is left to future work; (ii) rewiring is globally non-local—like-state interactions trigger formation of an edge to a uniformly random non-neighbor anywhere in the system, subject to a degree cap, rather than being restricted to neighbors of neighbors; (iii) we compare directly against a fixed-lattice control to assess the effect of topological freedom.

We find that adaptive rewiring strongly sharpens the finite-size transition relative to the fixed lattice: ρ_b jumps and per-seed distributions become bimodal at intermediate system sizes. Binder-cumulant trends, however, weaken with L rather than strengthening, consistent with a sharp continuous transition over the tested range and disfavoring a conventional first-order interpretation; a weakly first-order alternative with crossover length beyond $L = 384$ is not excluded. Control experiments indicate that some degree headroom is necessary for the sharp jump, though the effect is not cleanly separable from reinforcement-firing and formation-rate confounds.

The remainder of the paper is organized as follows. Section 2 defines the model. Sections 3 and 4 present results on the fixed lattice and adaptive network, respectively, including control experiments. Section 5 presents finite-size behavior near the sharp transition, including the Binder cumulant test. Section 6 discusses the mechanism and connections to related models.

2. Model

The model is a random-sequential interacting particle system with continuous internal states on an adaptive graph: nodes carry continuous $[0, 1]^3$ state vectors, updates are *asynchronous and random-sequential* (one randomly selected node updates at a time, with each update seeing the current configuration), and in the adaptive variant the topology coevolves with the dynamics. We refer to the dynamics as a "cellular automaton" loosely, following the convention in the lattice competition literature; in the strict sense the updates are not synchronous.

2.1. Fixed lattice

The substrate is an $L \times L$ two-dimensional torus: a square grid with periodic boundary conditions in both directions, giving every node exactly four neighbors. This 4-regular toroidal lattice serves as the fixed-topology control: periodic boundaries remove edge effects, so any difference between fixed-lattice and adaptive-network behavior is attributable to topological coevolution rather than to boundary geometry. The 4-regular degree also matches the initial mean degree of the adaptive network at $t = 0$, so divergence in late-time behavior reflects rewiring dynamics rather than initial-condition mismatch.

2.2. Node state and competition

Each node i carries a three-channel color value $\boldsymbol{c}_i = (r_i, g_i, b_i) \in [0,1]^3$. The three channels represent three competing species; the *dominant type* of node i is the channel with the largest value. Ties between channel values are uncommon: states are initialized continuously and then updated in increments of $\delta = 1/255$. When ties occur they are resolved by a deterministic implementation rule (documented in the public code). Because ties are rare and the three species are otherwise treated identically, the choice of tie-breaker does not affect the reported phase behavior within numerical resolution; the phase diagram is symmetric under species relabeling. The continuous color values endow each node with a "conviction": the *dominance margin* is the difference between the dominant channel and the maximum of the other two. A node with color $(0.78, 0.12, 0.04)$ is strongly red (margin 0.67), while $(0.35, 0.33, 0.31)$ is weakly red (margin 0.02) and easily converted.

At each elementary step a node i is chosen uniformly at random and interacts with each of its neighbors j. Channel values change in increments of $\delta = 1/255 \approx 0.004$:

- **Agreement** (same dominant type): both i and j increment their shared dominant channel by δ (clamped at 1).
- **Conflict** (different dominant types): the node with the smaller dominance margin shifts toward the winner's type: the winner's channel is incremented by δ and the loser's own dominant channel is decremented by δ. When margins are equal, the focal node i concedes.

A collective reinforcement rule amplifies local consensus: if at least $r_{\min}$ of i's neighbors share the same dominant type (as determined by the edge relationship after competition), that channel receives an additional increment. All experiments use $r_{\min} = 4$ unless otherwise noted.

One *frame* consists of $N = L^2$ elementary steps, corresponding to one sweep of the lattice in expectation.

2.3. Mutation

After node i has interacted with all its neighbors and reinforcement has been applied, i is reset to a uniformly random color—each channel drawn independently and uniformly from $[0,1]$—with probability μ. Mutation is applied once per node activation, not once per pairwise interaction. This mutation is the noise parameter that drives the phase transition: low μ favors local order; high μ disrupts it.

2.4. Adaptive topology

The adaptive variant begins from the same 4-regular toroidal lattice but allows the network to rewire. After each pairwise interaction between nodes i and j:

- If the edge is **compatible** (endpoints share dominant type after competition) and the focal node i has degree $< k_{\max}$: with probability $1/N$, i attempts to form a new edge to a non-neighbor. Up to 8 candidate indices are drawn uniformly at random from $\{0, \ldots, N-1\}$; a candidate is accepted if it is not i or j, is not already a neighbor of i, and itself has degree $< k_{\max}$. If none of the 8 candidates qualifies, the formation attempt is discarded. The new edge target is *not* required to share i's dominant type: compatibility-triggered formation selects *when* a node attempts to grow its neighborhood, and subsequent severance of incompatible edges then prunes the topology. Homophilic clustering therefore arises from compatibility-triggered formation plus selective survival, not from explicitly choosing same-type targets.

- If the edge is **incompatible** (endpoints differ): with probability $1/N$, the edge is severed, provided both endpoints retain degree ≥ 2.

The maximum degree is capped at $k_{\max} = 8$ (twice the initial lattice degree). This cap is not a passive parameter: changing $k_{\max}$ alters the equilibrium mean degree $\langle k \rangle$, which in turn changes the number of pairwise interactions per node activation, the reinforcement-firing probability under fixed $r_{\min}$, and the per-frame rewiring opportunity (Section 4.4). The minimum degree floor of 2 reduces the frequency of isolated nodes and extreme fragmentation, but does not guarantee global connectivity.

The rewiring probability is $1/N$ per pairwise interaction, and each frame contains $O(N\langle k \rangle)$ pairwise interactions ($\sim N$ activations $\times$ $O(\langle k \rangle)$ neighbors each), so the expected number of rewiring events per frame is $O(\langle k \rangle)$, *independent of N for bounded degree.* Extensive topological relaxation therefore requires $O(N)$ frames.

Each run is initialized from a 64-bit seed; the seed deterministically sets the initial colors and the state of the pseudo-random number generator used for all subsequent uniform draws (node selection, mutation, candidate sampling, severance acceptance). Two runs with the same seed and parameters produce identical trajectories.

In the ordered phase ($\mu < \mu_c$), nodes cluster near the degree cap: at $\mu = 0.33$, approximately 82% of nodes have $k = 8$ and 95% have $k \geq 7$. In the disordered phase ($\mu > \mu_c$), the distribution shifts toward the floor: at $\mu = 0.50$, approximately 44% of nodes have $k = 2$ and 71% have $k \leq 3$. The degree distribution is thus strongly phase-dependent (Fig. 2, Table 3).

The result is that ordered regions recruit long-range connections while disordered boundaries shed them. The topology and dynamics coevolve: the network is simultaneously the arena for competition and a product of it. A control variant restricts new-edge formation to nodes within graph distance 2 of the focal node while keeping all other rules identical (Section 4.4).

2.5. Algorithm summary

Table 1 gives the per-node update. One frame repeats this N times with uniformly random node selection.

Table 1: Per-node update for node i.

```
Snapshot neighbor list N(i)
for each j ∈ N(i):
    if same dominant type:
        increment shared channel on both i and j
        [adaptive] with prob. 1/N: attempt to form edge
        from i to random non-neighbor (both endpoints
        must have degree < k_max)
    else:
        weaker node shifts one unit toward winner
        [adaptive] with prob. 1/N: sever edge i–j
        if both keep degree ≥ 2
Reinforce: if ≥ r_min edges agree, boost that type
Mutate: with prob. μ, reset c_i to random
```

2.6. Observables

The primary order parameter is the *boundary density* ρ_b: the fraction of edges connecting nodes of different dominant types. Low ρ_b ($\lesssim 0.15$) indicates species-segregated order; high ρ_b ($\gtrsim 0.5$) indicates a well-mixed disordered state.

On the adaptive network, the mean degree $\langle k \rangle$ serves as a second, coupled order parameter: $\langle k \rangle \approx 7$–8 in the ordered phase (near the cap) and $\langle k \rangle \approx 3$–4 in the disordered phase (near the floor).

All reported values are time-averages over the second half of each run (discarding transients). Error bars on mean quantities in figures show the standard error of the mean ($\sigma/\sqrt{n}$) across independent seeds. Tables report the seed-to-seed standard deviation σ and population variance. The primary experiments use 64 seeds; the Binder cumulant analysis uses 64 seeds per μ point at every system size, including $L = 384$.

2.7. Note on the reinforcement threshold

On the fixed 4-regular lattice, $r_{\min} = 4$ requires all four neighbors to agree before reinforcement fires—effectively a unanimity rule. On the adaptive network, where degree can reach $k_{\max} = 8$, the same $r_{\min} = 4$ requires only half the neighbors to agree. This asymmetry is inherent to applying a fixed threshold across variable-degree topologies. We address its consequences with a control experiment (Section 4.5) showing that varying $r_{\min}$ on the fixed lattice does not change the transition character.

3. Fixed lattice: smooth crossover at $L = 128$

On the fixed 4-regular torus, at the tested system size ($L = 128$, 64 seeds, 50 000 frames), boundary density decreases smoothly with decreasing mutation probability (Fig. 1, Table 2). No discontinuity is observed at any μ tested: ρ_b ranges from 0.48 at $\mu = 0.33$ to 0.08 at $\mu = 0.10$, with a smooth crossover near $\mu \approx 0.20$. This section serves as a baseline: the tested fixed-lattice behavior at $L = 128$ is smooth, and Section 4 asks what changes under adaptive rewiring. We do not claim this as a thermodynamic statement about the fixed-lattice transition; multi-size fixed-lattice scaling is left to future work.

4. Adaptive network: sharp finite-size transition

Allowing edges to coevolve qualitatively changes the finite-size transition relative to the fixed-lattice baseline.

Table 2: Fixed lattice phase diagram (128×128, $r_{\min} = 4$, 64 seeds, 50k frames).

μ	$\bar{\rho}_b$	$\sigma(\rho_b)$	$\mathrm{var}(\rho_b)$
0.1000	0.0846	0.0059	3.42e-05
0.1111	0.0958	0.0074	5.42e-05
0.1250	0.1076	0.0066	4.31e-05
0.1429	0.1255	0.0035	1.20e-05
0.1667	0.1577	0.0036	1.29e-05
0.2000	0.2284	0.0016	2.66e-06
0.2500	0.3883	0.0012	1.39e-06
0.3333	0.4766	0.0008	7.04e-07

4.1. Sharp finite-size jump in coupled order parameters

The phase diagram of the adaptive network (Fig. 2, Table 3) shows both order parameters changing sharply between $\mu = 0.35$ and $\mu = 0.355$. At $\mu = 0.35$, the majority of seeds converge to the ordered phase ($\rho_b \approx 0.17$, $\langle k \rangle \approx 6.0$); at $\mu = 0.355$, the per-seed distribution is bimodal (see Section 4.2), and the reported seed mean $\rho_b \approx 0.48$, $\langle k \rangle \approx 4.2$ is therefore not a typical single-seed value but the mean over a two-basin distribution. By $\mu = 0.36$, all seeds are fully disordered ($\rho_b = 0.55$, $\langle k \rangle = 3.9$). The boundary density increases by a factor of $\sim 3\times$ and the mean degree drops by $\sim 1.5\times$ across this narrow window.

4.2. Topology-independent confirmation and finite-size coexistence

Because ρ_b depends on the edge set (both numerator and denominator), a sharp jump in ρ_b alone could in principle be a topological artifact of changing connectivity rather than of the node states. We rule this out with a state-only observable: the dominant-species fraction $f_{\mathrm{dom}} = \max(n_R, n_G, n_B)/N$ (Fig. 3), which drops sharply from ≈ 0.9 to $\approx 1/3$ across the same window—identical jump location, no edge set in the definition—confirming the transition reflects a genuine state change.

Histograms of per-seed ρ_b across the transition (Fig. 4) reveal the distribution shape more precisely than the scatter plots above. At $\mu = 0.33$, the distribution is a tight unimodal peak near $\rho_b \approx 0.10$. At $\mu = 0.345$, it remains unimodal but broadens. At $\mu = 0.35$, the distribution develops a long right tail, with most seeds near $\rho_b \approx 0.16$ but a few reaching $\rho_b > 0.3$. At $\mu = 0.355$, the distribution is *clearly bimodal*: one cluster near $\rho_b \approx 0.15$

Table 3: Adaptive network phase diagram (128×128, $r_{\min} = 4$, $k_{\max} = 8$, 64 seeds, 30k frames).

μ	$\bar{\rho}_b$	$\sigma(\rho_b)$	$\mathrm{var}(\rho_b)$	$\langle k \rangle$
0.1111	0.0270	0.0002	5.82e-08	7.90
0.1250	0.0306	0.0002	5.31e-08	7.89
0.1429	0.0352	0.0002	6.23e-08	7.88
0.1667	0.0418	0.0003	7.62e-08	7.87
0.2000	0.0511	0.0003	1.01e-07	7.85
0.2500	0.0665	0.0004	1.26e-07	7.80
0.3300	0.1016	0.0012	1.36e-06	7.35
0.3333	0.1046	0.0016	2.51e-06	7.29
0.3400	0.1131	0.0034	1.18e-05	7.06
0.3450	0.1257	0.0093	8.59e-05	6.75
0.3500	0.1748	0.0566	3.20e-03	5.97
0.3550	0.4779	0.1230	1.51e-02	4.21
0.3600	0.5483	0.0005	2.89e-07	3.90
0.3700	0.5511	0.0007	4.34e-07	3.87
0.4000	0.5589	0.0007	5.03e-07	3.79
0.4500	0.5706	0.0007	5.25e-07	3.68
0.5000	0.5813	0.0007	5.41e-07	3.59
1.0000	0.6581	0.0006	3.66e-07	3.12

and a second near $\rho_b \approx 0.55$, separated by a gap with no seeds. By $\mu = 0.37$, all seeds have collapsed to a tight unimodal peak in the disordered phase.

The bimodality at $\mu = 0.355$—with two well-separated peaks and an empty gap—is the strongest coexistence-like signature at $L = 128$. Section 5.1 shows that these signatures weaken with increasing system size.

4.3. Variance spike and initial-condition dependence

The *seed-to-seed variance of the per-run second-half time-averaged* ρ_b peaks sharply at $\mu = 0.355$, with $\mathrm{var}(\rho_b) = 1.5 \times 10^{-2}$, against an off-transition background of $\mathrm{var}(\rho_b) \sim 10^{-7}$ in both phases (Table 3; Fig. 2, inset). The off-transition background reflects tight single-basin sampling within seeds; the peak reflects seeds splitting between two basins. We use the spike to *localize* the transition, not as a diagnostic of its order: a peak in seed-to-seed variance occurs at both continuous and first-order transitions,

Table 4: Initial-condition dependence across the bistable window (128×128, 64 seeds, 30k frames).

μ	Init	$\bar{\rho}_b$	$\sigma(\rho_b)$	$\langle k \rangle$
0.340	random	0.1131	0.0034	7.06
	ordered	0.1133	0.0036	7.06
0.345	random	0.1257	0.0093	6.75
	ordered	0.1238	0.0075	6.80
0.350	random	0.1748	0.0566	5.97
	ordered	0.1778	0.0515	5.95
0.355	random	0.4779	0.1230	4.21
	ordered	0.4458	0.1438	4.36
0.360	random	0.5483	0.0005	3.90
	ordered	0.5482	0.0007	3.90

so the magnitude of the ratio to the off-transition value reflects the basin separation rather than the transition order.

To test for initial-condition dependence, we run at $\mu = 0.34$, 0.35, and 0.36 from both random and spatially ordered initial conditions (64 seeds each; Table 4). At $\mu = 0.34$, both conditions converge to order; at $\mu = 0.36$, both converge to disorder. At $\mu = 0.35$, a small fraction of seeds escape to the opposite basin (3/64 random-init, 2/64 ordered-init), with means close in both cases ($\bar{\rho}_b = 0.175$ vs. 0.178). This is weak initial-condition dependence rather than strong hysteresis.

4.4. Controls

To isolate which features of the rewiring drive the sharpening, we ran two families of controls: a local-only control restricting new edges to distance-2 neighbors (all other rules identical), and a degree-cap sweep varying $k_{\max}$.

Figure 5 compares four systems at $L = 128$ (64 seeds, 30 000 frames): the fixed lattice, the local-only adaptive network ($k_{\max} = 8$), the global adaptive network with $k_{\max} = 4$ (matching the fixed-lattice degree), and the standard global adaptive network with $k_{\max} = 8$.

On the local-only network, ρ_b rises smoothly from ≈ 0.13 at $\mu = 0.33$ to ≈ 0.48 at $\mu = 0.50$, with no sharp jump at any μ tested. The restriction removes the sharp jump, but it also shrinks the candidate non-neighbor pool

Table 5: Finite-size scaling at $\mu = 0.35$ (32 seeds per condition).

L	Init	$\bar{\rho}_b$	$\sigma(\rho_b)$	$\langle k \rangle$
64	random	0.1004	0.0008	7.80
	ordered	0.1006	0.0009	7.80
128	random	0.1819	0.0705	5.92
	ordered	0.1700	0.0500	6.07
256	random	0.4637	0.1151	4.21
	ordered	0.4517	0.1246	4.25

from $O(N) \approx 16{,}000$ system-wide to $O(k^2) \approx 50$–60 within graph distance 2 at $k_{\max} = 8$, so it does not isolate non-locality from formation rate.

A second family of controls varies the degree cap. At $k_{\max} = 4$ on the global adaptive network (matching the fixed-lattice degree), the sharp finite-size jump vanishes: ρ_b rises smoothly from ≈ 0.15 ($\mu = 0.15$) to ≈ 0.56 ($\mu = 0.50$), with seed-to-seed variance three orders of magnitude below the $k_{\max} = 8$ case. Because the network starts at degree $4 = k_{\max}$, there is no headroom for edge accumulation: at initialization every node is already at the cap, so formation attempts fail until severance opens a slot, and mean degree drops monotonically to $\langle k \rangle \approx 3.0$–3.7.

To disentangle headroom from outright formation suppression, we ran intermediate-headroom controls at $k_{\max} = 5$ and $k_{\max} = 6$ (one and two units of headroom respectively). Both recover a sharp coexistence-like jump (Fig. 5): at $k_{\max} = 5$, ρ_b jumps from ≈ 0.20 at $\mu = 0.27$ to ≈ 0.51 at $\mu = 0.30$; at $k_{\max} = 6$, the jump shifts to between $\mu = 0.30$ and $\mu = 0.33$ (from $\rho_b \approx 0.14$ to ≈ 0.53); at $k_{\max} = 8$, the jump occurs near $\mu \approx 0.345$. The transition location drifts monotonically toward higher μ with increasing headroom. Over the tested values, the sharp jump is absent at $k_{\max} = 4$ and present for $k_{\max} \geq 5$, suggesting a threshold-like onset of the sharp coexistence-like behavior once some degree headroom is available.

Together with the fixed-lattice baseline and the local-only control, this $k_{\max}$ sweep is *consistent with* a headroom requirement: some room for degree to accumulate in ordered regions appears necessary for the sharp jump, though the full $k_{\max} = 8$ headroom of the standard setting is not. The reading is not, however, an unconfounded measurement of headroom alone. Three coupled effects vary together with $k_{\max}$: (i) the equilibrium ordered-

phase mean degree changes ($\langle k \rangle \approx 3.1, 4.8, 5.8, 7.9$ at $k_{\max} = 4, 5, 6, 8$), so the per-activation interaction count varies; (ii) the reinforcement-firing probability under fixed $r_{\min} = 4$ varies with $\langle k \rangle$ (it is structurally suppressed at $k_{\max} = 4$ where $\langle k \rangle < r_{\min}$, and the effective ratio $r_{\min}/\langle k \rangle$ varies from $\approx$0.83 to 0.51 across $k_{\max} \in \{5, 6, 8\}$); and (iii) the per-frame rewiring opportunity scales with $\langle k \rangle$. A monotonic shift of the jump location with $k_{\max}$ is therefore consistent with several mechanisms beyond pure topological headroom. The local-only control is similarly confounded by candidate-pool shrinkage from $O(N)$ to $O(k^2)$. Cleanly isolating any one of these factors would require an $r_{\min}$ sweep on the adaptive network at fixed $k_{\max} = 8$ and a matched-formation-rate non-local control, both left to future work.

4.5. Reinforcement-threshold control

To probe an effective-threshold artifact ($r_{\min} = 4$ is unanimity at degree 4 but only a 50% majority at degree 8), we repeated the fixed-lattice experiment with $r_{\min} = 2$ (64 seeds, 50 000 frames), matching the fractional threshold of $r_{\min} = 4$ at degree 8. The $L = 128$ fixed-lattice curve remains smooth under this match, and ρ_b values shift downward only slightly. The smooth fixed-lattice behavior therefore survives a fractional-threshold match to the adaptive network, so the fixed-lattice/adaptive contrast is not produced by the threshold mismatch alone. A separate confound—the dynamic variation of $r_{\min}/\langle k \rangle$ with phase on the adaptive network—is not addressed here and is discussed in Section 6.1.

5. Finite-size behavior near the sharp transition

This section presents two complementary finite-size scans: a bistability scan at fixed $\mu = 0.35$ across $L \in \{64, 128, 256\}$ to show that seeds split between basins at accessible sizes (this subsection), and a denser Binder-cumulant scan across a finer μ grid with twice the seed count to test how this finite-size signature evolves with system size (Section 5.1).

Bistability at fixed μ. We measure ρ_b at $\mu = 0.35$ for $L = 64$, 128, and 256, with both random and ordered initial conditions (32 seeds each; Table 5).

At $L = 64$, all 64 seeds converge tightly to the ordered phase ($\rho_b \approx 0.10$, $\langle k \rangle \approx 7.8$, combined $\mathrm{var}(\rho_b) = 7.1 \times 10^{-7}$). At $L = 128$, seeds split between basins with $\mathrm{var}(\rho_b) = 3.8 \times 10^{-3}$. At $L = 256$ (100 000 frames for equilibration), most seeds have crossed to the disordered side ($\bar{\rho}_b \approx 0.46$) but

variance remains high ($\mathrm{var}(\rho_b) = 1.4 \times 10^{-2}$), indicating that the system is still bistable at this size.

The combined variance *increases* by four orders of magnitude from $L = 64$ to $L = 256$. At $\mu = 0.35$ the pseudocritical region drifts toward this value with increasing L (Section 5.1), so part of the variance increase reflects the approach to the pseudocritical point. Nonetheless, the strong broadening and bimodal splitting at $L = 128$ and 256 indicate coexistence-like behavior at accessible system sizes.

5.1. Binder cumulant

The Binder analysis is a separate, denser scan than the bistability check above: a fine μ grid ($\Delta\mu = 0.001$) with 64 seeds per (L, μ) point—twice the per-condition seed count of the bistability scan—at $L \in \{64, 128, 256, 384\}$. We compute the Binder cumulant $U_L = 1 - \langle \rho_b^4 \rangle / (3 \langle \rho_b^2 \rangle^2)$ from the second-half time-averaged ρ_b value of each independent seed (one value per seed), with the moments $\langle \cdot \rangle$ taken across the seed ensemble. For a tight single-peaked distribution $U_L \to 2/3$; when the distribution is bimodal, U_L develops a dip below 2/3. For a conventional first-order transition the bimodal peaks survive $L \to \infty$, so $U_{\min}$ should remain below 2/3 (the bounded support $\rho_b \in [0, 1]$ means it need not diverge as in the unbounded-magnetization case, but it should not recover toward the single-peak value). An upward trend toward 2/3 over the size range where the trend is measured is therefore inconsistent with first-order behavior.

Caveats specific to a one-sided order parameter. Standard Binder analysis is formulated for a symmetric order parameter (e.g., Ising magnetization, where the disordered phase has $\langle m \rangle = 0$ and the ordered phase has $|m| \neq 0$, so U_L takes different limits in the two phases and its curves cross near μ_c). Here $\rho_b \in [0, 1]$ is one-sided: it has nonzero mean in both phases, so a tight ordered single peak ($\rho_b \approx 0.1$) and a tight disordered single peak ($\rho_b \approx 0.55$) both give $U_L \to 2/3$. Only bimodality drives U_L below 2/3. Consequently the diagnostic content of U_L in this paper is the dip *depth* as a function of L, not a universal-value crossing of U_L curves at μ_c, which the standard analysis is not set up to deliver here.

Figure 6 shows U_L vs. μ for $L = 64$, 128, 256, and 384. Run lengths are 30 000–50 000 frames at $L \in \{64, 128\}$, 100 000 frames at $L = 256$, and 200 000 frames at $L = 384$. For $L \in \{128, 256\}$ this roughly tracks L^2 scaling,

and qualitative time-series spot-checks (Supplementary Material, Sec. S3) suggest comparable coverage of τ_{eq} at the two sizes.[1]

At $L = 384$ run lengths fall short of the same scaling: matching the per-τ_{eq} coverage of $L = 256$ under a plausible dynamical exponent $z \gtrsim 2$ would require $\sim$225 000 frames or more, so the 200 000-frame $L = 384$ runs cover fewer autocorrelation times than the smaller-size runs, and time-series spot-checks confirm within-run basin-switching similar to $L = 256$ (Supplementary Material, Sec. S3). The $L = 384$ point should therefore be treated as supporting but not as load-bearing: incomplete sampling of basin residence times can bias the per-seed Binder estimate, and the direction of that bias depends on the balance between trapped single-basin trajectories (which push $U_{\min}$ toward 2/3) and trajectories that switch during the averaging window (which can push it the other way). The trend that drives the conclusions below is therefore the $L = 128 \to 256$ comparison, where run lengths more nearly track the expected L^2-like scaling.

The *position* of the minimum shifts to lower μ with increasing L: $\mu^*(64) \approx 0.360$, $\mu^*(128) \approx 0.352$, $\mu^*(256) \approx 0.349$, $\mu^*(384) \approx 0.348$. The drift slows with increasing L, consistent with convergence toward a finite limiting value, though four points do not permit a reliable extrapolation. The measured dip values are (bootstrap 95% CI in brackets, 64 seeds at each size): $U_{\min} = -0.39$ $[-0.85, -0.05]$ ($L = 64$), 0.26 $[0.18, 0.36]$ ($L = 128$), 0.52 $[0.49, 0.55]$ ($L = 256$), and 0.58 $[0.56, 0.61]$ ($L = 384$). Confidence intervals are computed by resampling per-seed ρ_b values with replacement (seed-level non-parametric bootstrap, 2000 resamples; $U_{\min}$ recomputed on each resample and the 2.5/97.5 percentiles reported). The load-bearing finding is the $L = 128 \to 256$ change: $U_{\min}$ moves from 0.26 to 0.52 (CIs non-overlapping), a substantial fraction of the way to the single-peak value 2/3 at run lengths that roughly satisfy L^2 scaling. This is inconsistent with a first-order Binder dip, which should not approach the single-peak value at larger L. The $L = 64$ point lies in the same direction but the system contains only 4096 nodes; small-system corrections plausibly dominate, and it is consistent with the $L = 128 \to 256$ trend only if those corrections are accepted. The $L = 384$ point is also consistent with the trend but, as discussed above, is downgraded given the equilibration limitation. Taken together, the data

[1] A quantitative integrated-autocorrelation-time measurement of ρ_b vs. L would put this on a firmer footing and is part of the future-work list (Section 6).

disfavor a conventional first-order interpretation over the tested sizes and are consistent with a sharp continuous transition. We do not claim to have established a continuous transition: a weakly first-order transition with a Binder crossover length beyond $L = 384$ would mimic the same finite-size signature, and settling the asymptotic order would require systems substantially beyond $L = 384$ and longer run lengths.

Per-seed ρ_b distributions at $L = 384$ (Supplementary Material, Sec. S2) show that bimodality persists at the largest size but is confined to a narrow window $\mu \in [0.347, 0.349]$ around the Binder minimum—substantially narrower than the $L = 128$ bimodal window of Figure 4, consistent with the same sharpening-yet-continuous trend as the Binder recovery toward 2/3.

As a consistency check we attempted a two-parameter Binder collapse $U_L(\mu) = \tilde{U}((\mu - \mu_c)L^{1/\nu})$. The collapse fails for $L \in \{128, 256, 384\}$ (the $L = 128$ dip is too deep to be absorbed by any rescaling, consistent with $L = 128$ lying outside the leading-order finite-size scaling regime). Restricting to two curves $L \in \{256, 384\}$ yields a tight collapse, but two curves with two free parameters provide essentially no constraint on the exponent (bootstrap 95% CI $1/\nu \in [0.97, 1.99]$ spans the standard two-dimensional continuous-transition exponents without selecting one). We therefore do not assign a universality class from this analysis and treat the collapse only as a consistency check; see Supplementary Material, Sec. S1 for the full procedure and figure.

6. Discussion

6.1. Mechanistic interpretation

The control experiments in Section 4.4 narrow the set of topological ingredients that contribute to the sharp finite-size transition at $L = 128$. Sweeping $k_{\max} \in \{4, 5, 6, 8\}$ shows that any degree headroom (one unit or more) recovers the sharp coexistence-like jump, with the jump location shifting monotonically toward higher μ as headroom grows. The local-only control also restores smoothness but is confounded by candidate-pool shrinkage from $O(N)$ to $O(k^2)$ and reduced effective formation rate; non-local formation per se is implicated only indirectly. Whether the formation/severance asymmetry contributes beyond non-local formation alone has not been tested directly (the converse control, making severance also non-local, remains future work).

A second confound bounds the headroom interpretation. At $k_{\max} = 4$, mean degree in the ordered phase drops to $\langle k \rangle \approx 3.1$, below the reinforcement

threshold $r_{\min} = 4$; reinforcement firing is therefore physically suppressed by construction at $k_{\max} = 4$, not by absence of a homophily–topology feedback loop alone. The $k_{\max} = 4$ vs. $k_{\max} \geq 5$ contrast partly reflects this trivial reinforcement-firing-rate effect. A more interpretable signal is the monotonic μ-shift of the jump location across $k_{\max} \in \{5, 6, 8\}$, where $\langle k \rangle \approx 4.8, 5.8, 7.9$ in the ordered phase (all above $r_{\min}$) and the jump location moves from $\mu \approx 0.30$ to 0.345 with increasing headroom. The direction of this shift is consistent with the homophily–topology picture: more headroom yields larger ordered-phase $\langle k \rangle$, so each node experiences more reinforcement firings per frame in the ordered phase, making it more robust against mutation and shifting μ_c upward. Even within the $k_{\max} \geq 5$ regime the effective ratio $r_{\min}/k$ varies from ≈ 0.83 to 0.51, so the shift cannot be cleanly attributed to headroom alone; a definitive disentanglement would require an $r_{\min}$ sweep on the adaptive network at fixed $k_{\max} = 8$, which we leave to future work.

Interpretation: a homophily–topology feedback loop. We offer the following picture as an interpretation of the data above, not as an independently established mechanism. In the ordered phase, compatible neighbors trigger formation events that accumulate degree ($\langle k \rangle \approx 7$–$8$ at $k_{\max} = 8$), and the larger neighborhood provides both more reinforcement firings and more competition partners of the same type, stabilizing order against mutation; in the disordered phase, incompatibility drives severance, $\langle k \rangle$ falls toward the floor (≈ 3–4), and sparse connectivity makes spontaneous ordering difficult. At intermediate system sizes this positive feedback produces the coexistence-like signatures of Section 4; the Binder cumulant trend (Section 5.1) shows that these signatures weaken at larger L, indicating that any effective free-energy barrier between basins does not grow unboundedly with system size. This picture is qualitatively consistent with prior adaptive-network results in voter, SIS, and cyclic-dominance models [5, 6, 7, 9, 10, 8, 11], where positive feedback between state coherence and network structure has been identified as a generic sharpening mechanism. The loop predicts a directional, falsifiable signal: an $r_{\min}$ sweep at fixed $k_{\max} = 8$ (which decouples the reinforcement threshold from $\langle k \rangle$ without changing the topology dynamics) should shift μ_c *downward* as $r_{\min}$ rises—because fewer firings per frame in the ordered phase reduces the mutation rate the system can absorb before losing order—and *upward* as $r_{\min}$ falls. Testing this is left to future work.

6.2. Limitations and future work

Several aspects of this study would benefit from further investigation.

Equilibration near μ_c. Time-series spot-checks at $L = 256$ and $L = 384$ show that some seeds undergo within-run basin-switching during the averaging window, so second-half averages near the Binder minimum partly conflate the two basins (Supplementary Material, Sec. S3). The $L = 384$ run length is short of the L^z scaling that would match $L = 256$ coverage under a plausible $z \gtrsim 2$; this is the reason the $L = 384$ Binder value is treated as supporting rather than load-bearing (Section 5.1). A quantitative integrated-autocorrelation-time measurement of ρ_b vs. L and longer $L = 384$ runs would clarify both effects.

Statistical uncertainty in fourth-moment quantities. The 64-seed Binder analysis is on the low end for a cumulant dominated by tail behavior of the per-seed ρ_b distribution. The bootstrap CIs reported in Section 5.1 estimate sampling uncertainty conditional on the run length used; they do not capture equilibration uncertainty, because resampling partially-equilibrated samples preserves the equilibration bias. A higher seed count (e.g. ≥ 128) at $L = 384$ together with the longer runs above would tighten both sources of uncertainty.

Universality class. The two-parameter Binder collapse at $L \in \{256, 384\}$ is consistent with standard two-dimensional continuous-transition exponents but is weakly constrained by only two curves (Supplementary Material, Sec. S1). A definitive universality-class assignment would require simulations at $L \geq 512$ or correction-to-scaling fits with additional free parameters.

Mechanism isolation. The mechanism claim rests on ablation controls, not on a direct measurement of the feedback loop. The single highest-value missing control is an $r_{\min}$ sweep on the adaptive network at fixed $k_{\max} = 8$ (Section 6.1), which would disentangle degree headroom from the reinforcement-firing-rate effect. A matched-formation-rate non-local control and a non-local-severance control would further sharpen the roles of formation range and the formation/severance asymmetry, respectively.

Discrete-state ablation. The model uses a continuous dominance margin ("conviction-weighted" rule) for resolving conflicts. Whether this continuous margin is essential to the phenomenology, or whether a discrete three-state variant

(e.g. majority-update or voter-style conflict resolution) on the same adaptive graph reproduces the sharp finite-size behavior, has not been tested. A near-term followup is to run such a discrete-state variant under otherwise identical rules and parameters and compare the Binder trends.

Mean-field theory. A mean-field analysis for the coupled ρ_b–$\langle k \rangle$ dynamics would help anchor the simulation results and predict the location of the bistable window as a function of model parameters.

7. Conclusion

What was found. At $L = 128$ the fixed toroidal lattice shows a smooth crossover-like loss of order. Adaptive non-local rewiring sharpens the same local rules into a coexistence-like finite-size jump in coupled order parameters ($\rho_b : 0.17 \to 0.48$, $\langle k \rangle : 6.0 \to 4.2$ across $\mu \in [0.35, 0.355]$), bimodal per-seed distributions at intermediate system sizes, and a sharply localized seed-to-seed variance peak. Between $L = 128$ and $L = 256$ the Binder minimum moves from 0.26 to 0.52 (CIs non-overlapping, run lengths roughly satisfying L^2 scaling)—a substantial fraction of the way to the single-peak value 2/3. This is the load-bearing evidence: it disfavors a conventional first-order interpretation over the tested sizes and is consistent with a sharp continuous transition. A degree-cap sweep is consistent with a degree-headroom requirement, with the jump location drifting monotonically toward higher μ as headroom grows. The mechanism reading is consistent with the data but does not rest on an unconfounded measurement; the limits are listed next.

What remains open. We do not claim to have established a continuous transition. A weakly first-order alternative with crossover length beyond $L = 384$ would mimic the same finite-size signature, and the $L = 384$ Binder estimate is limited by fewer sampled autocorrelation times than the smaller-size runs. The mechanism is also not fully isolated: changing $k_{\max}$ changes degree headroom, reinforcement firing, interaction count, and rewiring opportunity simultaneously, while the local-only control also changes the candidate pool. Adaptive topology therefore emerges as a strong finite-size sharpening mechanism in this conviction-weighted competitive model, while the asymptotic order of the transition and the clean separation of rewiring ingredients remain open.

CRediT authorship contribution statement

Evan W. Martin: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code (C source for the random-sequential update kernel and adaptive-graph data structure), parameter files for every reported run, raw per-seed CSVs, and the Python analysis and plotting scripts used to produce every figure and table are publicly available at `https://github.com/evanwmart/competitive-ca` (tag corresponding to this submission: see repository README for the exact commit and a step-by-step recipe that reproduces the figures and tables from the raw data).

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Anthropic's Claude for manuscript revisions and bug-fixing assistance in supporting code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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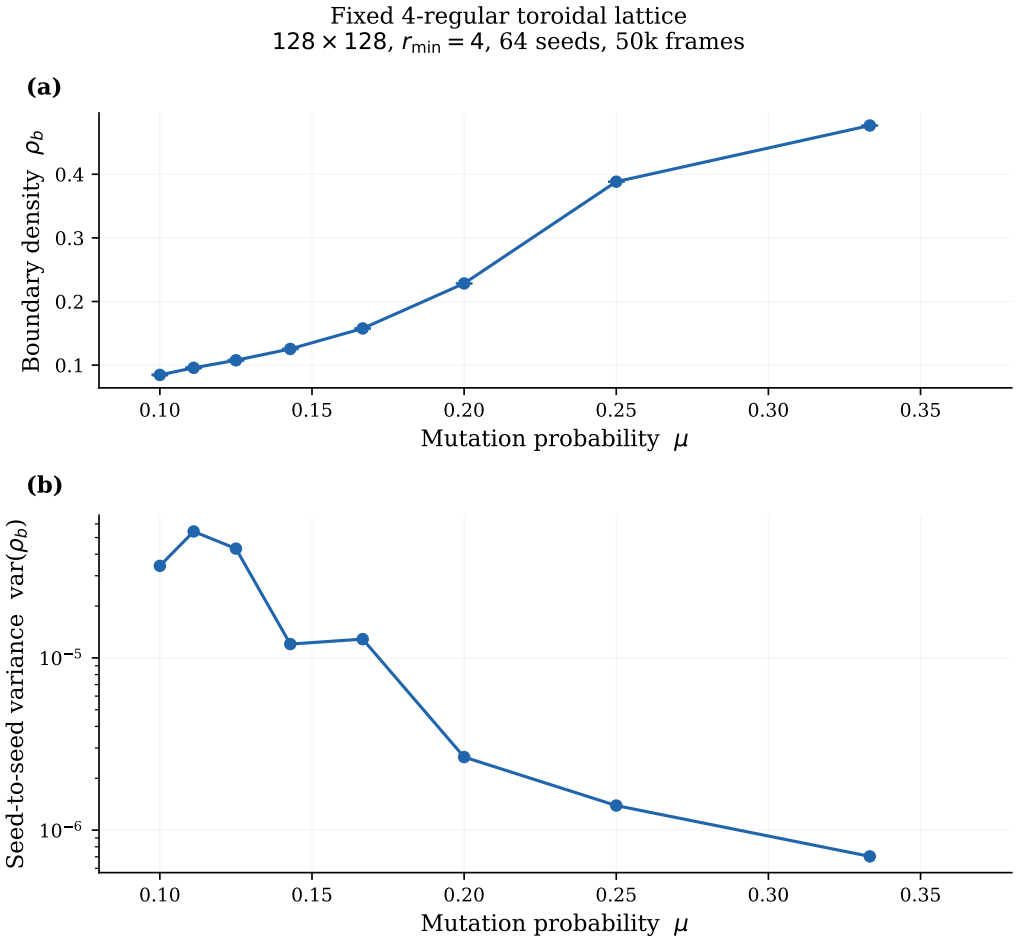

Figure 1: **Fixed lattice phase diagram** (128×128, $r_{\min} = 4$, 64 seeds, 50k frames). (a) Mean boundary density ρ_b vs. μ. The transition is smooth. (b) Seed-to-seed variance of ρ_b.

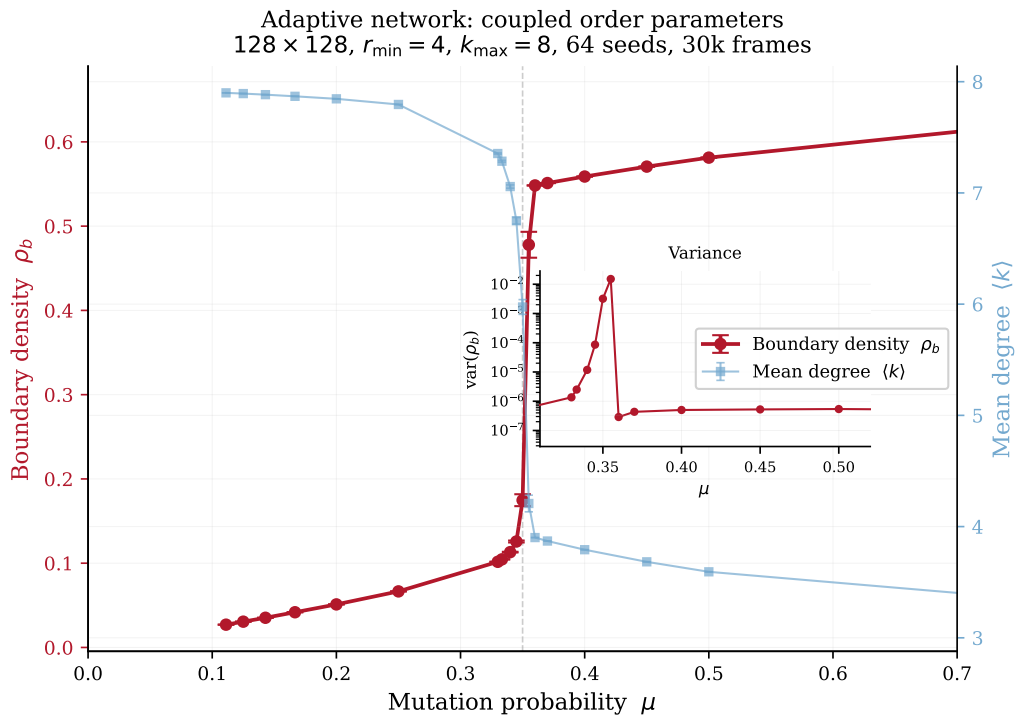

Figure 2: **Adaptive network phase diagram** (128×128, $r_{\min} = 4$, $k_{\max} = 8$, 64 seeds, 30k frames). Boundary density ρ_b (red, left axis) and mean degree $\langle k \rangle$ (blue, right axis) vs. mutation probability μ. Both order parameters change sharply in the window $\mu \in [0.35, 0.355]$. Within this window the per-seed distribution is bimodal (Fig. 4), so the plotted means are averages across two basins, not typical single-seed values; the distributional view in Fig. 4 is the more informative summary in this region.

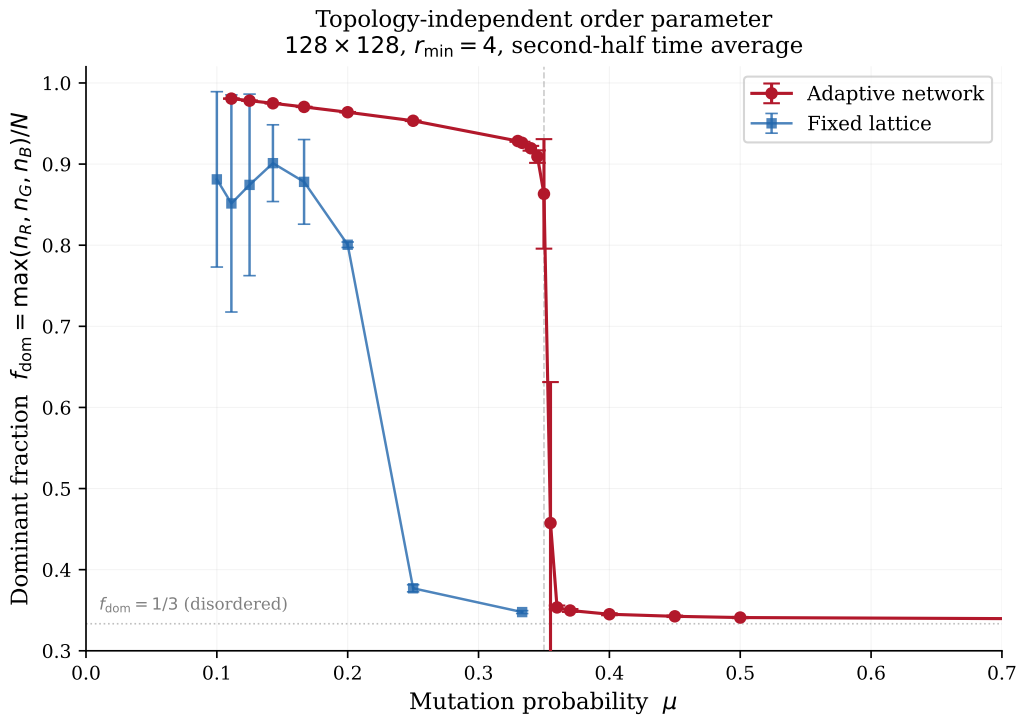

Figure 3: **Topology-independent order parameter.** f_{dom} vs. μ for the adaptive network (red) and fixed lattice (blue). The sharp drop indicates that the transition reflects a genuine state change. Dashed line: $f_{\text{dom}} = 1/3$ (disordered limit).

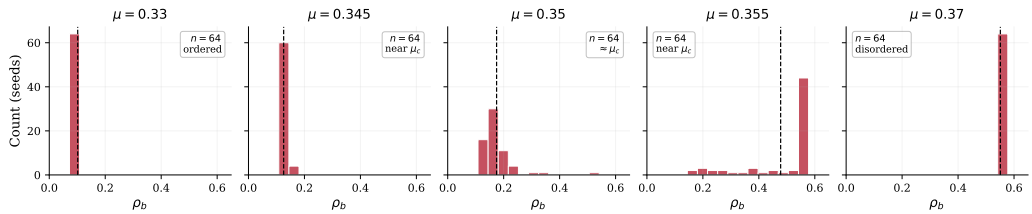

Figure 4: **Distribution of per-seed ρ_b across the finite-size transition at** $L = 128$ on the adaptive network (64 seeds, 30k frames). At $\mu = 0.355$, the distribution is clearly bimodal at this size, with an empty gap between ordered and disordered basins; this is the strongest coexistence-like signature at $L = 128$. Vertical dashed lines mark the seed mean (which, at this μ, sits between the basins and is not representative of any single seed). Some seeds near the transition may undergo basin switching during the averaging window (cf. Sec. S3).

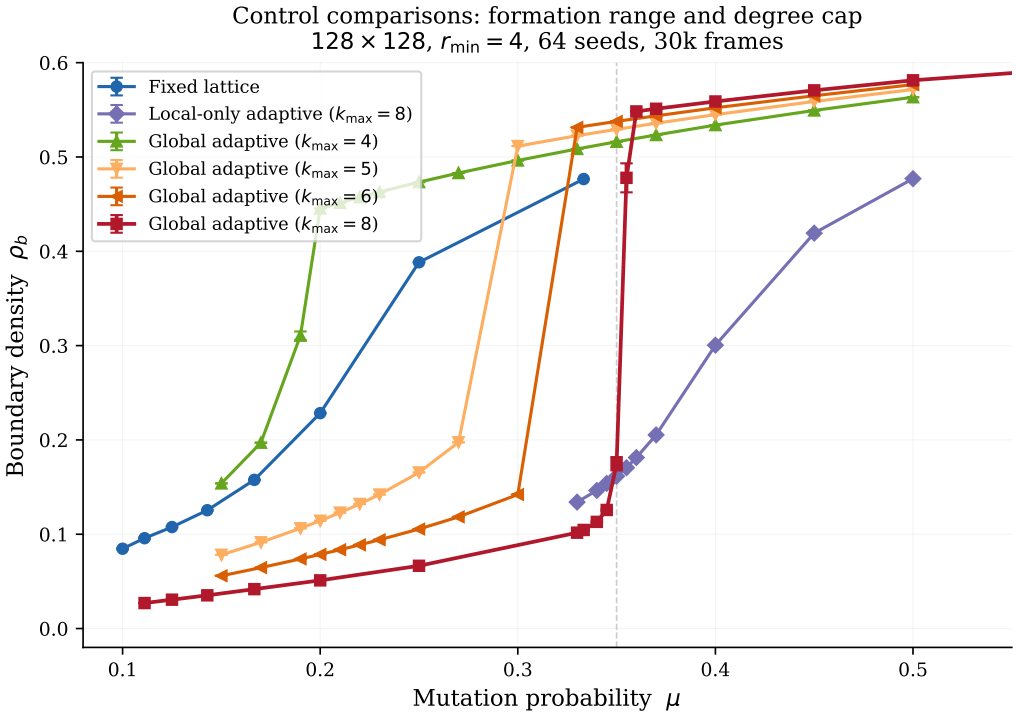

Figure 5: **Control comparisons: formation range and degree cap** (128×128, $r_{\min} = 4$, 64 seeds, 30k frames). Fixed lattice (blue), local-only adaptive with $k_{\max} = 8$ (purple), and global adaptive at four degree caps $k_{\max} \in \{4, 5, 6, 8\}$ (green, light orange, dark orange, red). The coexistence-like finite-size jump at $L = 128$ is absent at $k_{\max} = 4$ but present at every $k_{\max} \geq 5$; the jump location shifts to higher μ with increasing headroom. Caveats: at $k_{\max} = 4$ the mean ordered-phase degree drops below $r_{\min}$, so reinforcement firing is suppressed by construction (see Section 6.1); the local-only control shrinks the candidate pool and so does not cleanly isolate non-locality.

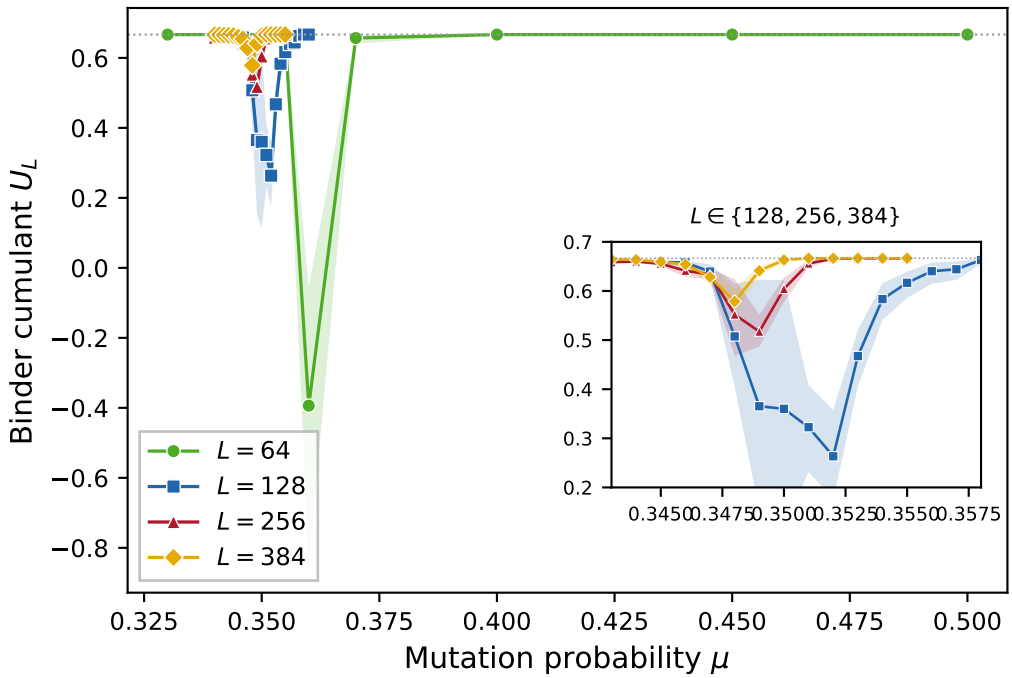

Figure 6: **Binder cumulant U_L vs. μ** for $L = 64$, 128, 256, and 384 on the adaptive network (fine μ grids, $\Delta\mu = 0.001$; 64 seeds per point). The dip shifts to lower μ with increasing L (pseudocritical drift). The primary finding is that $U_{\min}$ moves from 0.26 at $L = 128$ to 0.52 at $L = 256$ (CIs non-overlapping), a substantial fraction of the way to the single-peak value 2/3 and inconsistent with conventional first-order scaling. The $L = 64$ and $L = 384$ points are consistent with this trend but each carries a caveat (small-system corrections and equilibration respectively; see text).